Design Assignment #2:
Designing a Printed Circuit Board

In this module you will learn how to lay out part of a printed circuit board (PCB) in software. A PCB is a secure way to connect all of the components in an electronic circuit. While a breadboard can accomplish the same task, it has many limitations: parts can fall out, it is difficult or impossible to work with surface mount components, high frequency or high power voltages are not permitted, it is large and bulky, etc. A PCB is a way to custom build a small board to hold all of your components in place, via solder, and embed the wires, or traces, directly into the board. It is the way almost all circuits are constructed in practice. [Note that in electrical and computer engineering practice reliability is very important. If you design and sell unreliable widgets (whatever those may be) you not only are going to get a bad reputation, but will hurt the bottom line in lost sales and returns, and even open yourself up to lawsuits should critical parts fail.] You'll learn the PCB design tools used in ECE and you'll come up with your own unique design. Have fun with this! Think of a PCB like a canvas and it is your job as an artist (engineer) to use your brush (PCB design software) to move paint (components and traces) around the canvas to paint your picture (circuit). Two artists using the same brushes and paints can come up with vastly different interpretations of the same idea!

Because there is no one "right" way to lay out a PCB you will each come up with your own unique design, with different placement of components and different traces. [This is not to say there aren't better and worse ways to lay out a PCB, just that there isn't a single right answer. There are a lot of best practice techniques we'll be discussing throughout the course.] Because later course modules depend on this PCB you will eventually work with a common PCB design which will be provided to you later. [From a practical perspective it can take a few weeks to get a custom board designed, fabricated, and into your hands. We can't afford the time in the course for every person to design their own board and have it fabricated. Also this would significantly increase the cost.] As a result, this module will only go as far as laying out a PCB in software. You will not be required to fabricate your PCB. If you have interest in going that extra step and physically creating your own PCB we have the ability to send design files out to be professionally manufactured, like the PCBs you will use to build your DAMNED. We encourage you to speak with Matt if you are interested in fabricating your own PCB.

Next week you'll solder components onto the PCB we had professionally fabricated to begin to move towards a hardware implementation of the DAMNED project.

Objectives

- Understand the way PCB's are fabricated and laid out and some best practices in designing them.
- Conceptually follow the link between how a schematic is translated to a printed circuit board.
- Gain familiarity with KiCad EDA software.
- Understand signal flow between, and the types of, layers on a PCB.

- Design a valid PCB by placing all components on the board, laying connecting traces between components, eliminating all ratsnest connections, and avoiding any Design Rule Check (DRC) errors

Materials Needed

- KiCad software (available on all ECE computers in Dana 303, 305, 307, as well as the Maker-E. Also available for free from the KiCad website for Windows/Mac/Linux. See below for details.)
- Complete schematic (PDF)
- Partial KiCad schematic
- KiCad PCB skeleton

What you need to Know from Moodle Readings

- Understand the basics of PCBs
- How to use the KiCad EDA software
- How we design PCBs at Bucknell

More Resources that may be Helpful

- KiCad PCB layout in 5 steps (video)
- Importing Makerstock libraries
- KiCad 10.0 getting started
- KiCad 10.0 PCB Editor manual
- KiCad library conventions
- KiCad user forums
- KiCad Discord channel

Relation to the DAMNED Project

This design assignment will teach you how to design a printed circuit board or PCB. PCBs are the easiest way to fabricate professional quality circuits, and there are many companies that will accept designs as Gerber files, which are a particular format for laying out circuit boards. Once you have this skill down, you will know the basics of creating just about any circuit from a schematic. You will need a PCB to hold all of the electronic components for your DAMNED device.

What to Do and Know

The material below outlines what you should learn and gain skills in by the end of this module as well as what to include and discuss in the write up for the design assignment. It is important to realize that learning isn't binary - that is you know something or you don't, or you can do something or you can't - rather knowledge and skill improves over time as you gain experience and wisdom. Knowledge and skill must be practiced to improve. Consider this project an opportunity to gain practice modeling the behavior of a circuit. Some of you may already be experienced at what this module is asking you to learn and do, others may be encountering it for the first time. If you already know this, teach others. If this is new to you, realize the knowledge and skills will become better over time

Last week you created a schematic of the voltage regulator circuit and modeled it to determine how well it performed. This week your task is to place those components onto a printed circuit board. Each component--resistors, capacitors, integrated circuits, etc.-- may come in many different shapes and sizes. The physical realization of a component is known as a package. A single component or device (for example, a 555 timer) may come in multiple different packages (see pages 22-27 of the 555 datasheet). The area that a package occupies on a PCB is known as a **footprint** (or sometimes land pattern). While the footprint doesn't matter when you abstract a circuit to a schematic to model it, it **does** matter when you create a PCB since the component has to physically align with the holes and pads it will be attached to. So the first step is to go back and ensure that all components in your schematic have correct footprints. The correct footprints are those that match the footprints for the components we purchased and are detailed in the datasheets for these components. [Note that the cost of components drops significantly the more you order, so rather than have many different resistor footprints available we try to standardize on one size, in our case an 0805 which is 0.08 inches by 0.05 inches. Such a resistor might cost 10¢ if I buy just one, but 0.66¢ if I buy 1000.]

[There are many different pieces of electronic design automation (EDA) software for capturing schematics and laying out PCBs. We use KiCad because it is easy to learn with lots of available videos, guides, references, and crucially a large library of component models and footprints. In industry you are more likely to see software like Altium Designer or Cadence Orcad which have steep learning curves. But the basic concepts are the same between all software so getting a good foundation here will allow you to easily transfer skills to different software.] Begin by following the guide listed above in the **More Resources that may be Helpful** section to import the Makerstock component libraries into your instance of KiCad. Now download the zip file from the Design Assignment 2 section of Moodle. Create a new DA2 folder inside of your course Google Drive folder and unzip the contents of the zip file into the DA2 folder. There should be four KiCad files: one project, two schematics, one PCB.

With those files extracted, double click the project file (file extension .kicad_pro) to open the project. Always launch the KiCad project file when you are working with a schematic and PCB. That way the two stay linked together. **Failure to open a project file, and instead work on the schematic and PCB as individual files outside of a project, will lead to discontinuities between the two and cause problems!**

Schematic Editor

With the project open, now launch the Schematic Editor to display the DA2 schematic. You should recreate your voltage regulator circuit from Design Assignment 1 in this DA2 schematic. When recreating your circuit in the designated Power area of the KiCad schematic you can skip the 5V DC source (as well as the sinusoidal source) and load resistors you used in DA1. Those were only necessary when simulating the regulator. These devices are not necessary when creating a circuit which will ultimately become a PCB. Figure 1 below shows you what you can ignore from your DA1 circuit.

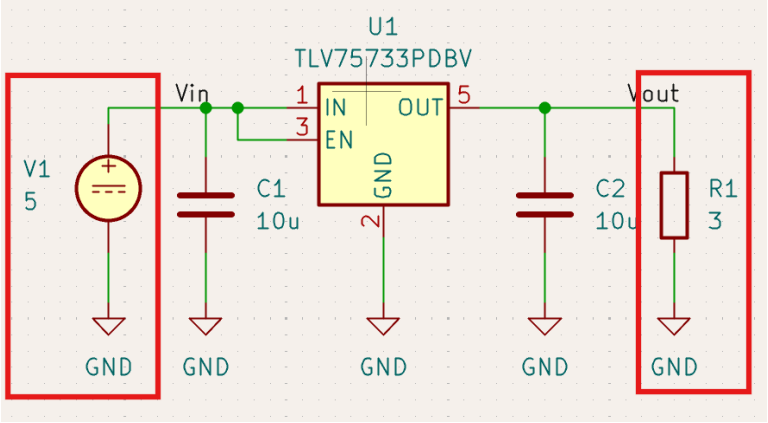


Figure 1: Do not add the DC voltage source and load resistor (boxed in red) when working on DA2

Notice that within the DA2 schematic you have been given, the components are grouped into modules or blocks. Your work is to recreate your regulator circuit from DA1 within the block labeled, “Power”.

You may also notice several green wires (lines) with black text above them (USB_D+, RESET, BOOT, etc.) in the schematic. These are labels.

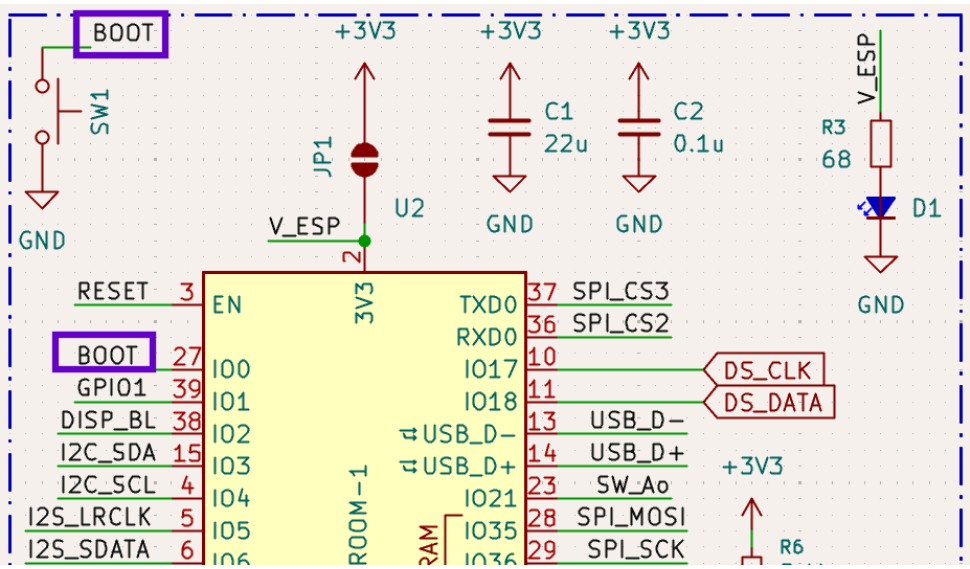


Figure 2: Example of net labels. Note the black BOOT label (purple box) connected to the top of switch SW1. This pin is also connected (virtually) to pin 27 (purple box) of device U2 (ESP32-S3-WROOM-1) below.

Any two points that have the same label name are virtually connected. This is a convenient way to make connections without having distracting wires running all over your screen. You will also notice that we have broken the design up into modules as designated by blue boxes. Note that labels are **not** simply text! If you place text on your schematic (appears blue) you will make no

connection to any nets. You **must** place a label (Place menu -> Add label) in order to make a connection to a net within your circuit.

In order to connect the input of your voltage regulator circuit to the rest of the DAMNED schematic you will need to add a special power symbol, Vcc, to the input of your circuit. This symbol is provided for you already in the given schematic and you can simply copy and paste one. For future designs you can also add other power connectors like +3V3 and GND by pressing the **P** key on your keyboard or copy/pasting existing symbols. The example below shows the completed circuit which you have not yet drawn. For now you can simply copy and paste a VCC connector in this area and then connect it after you complete the rest of the circuit.

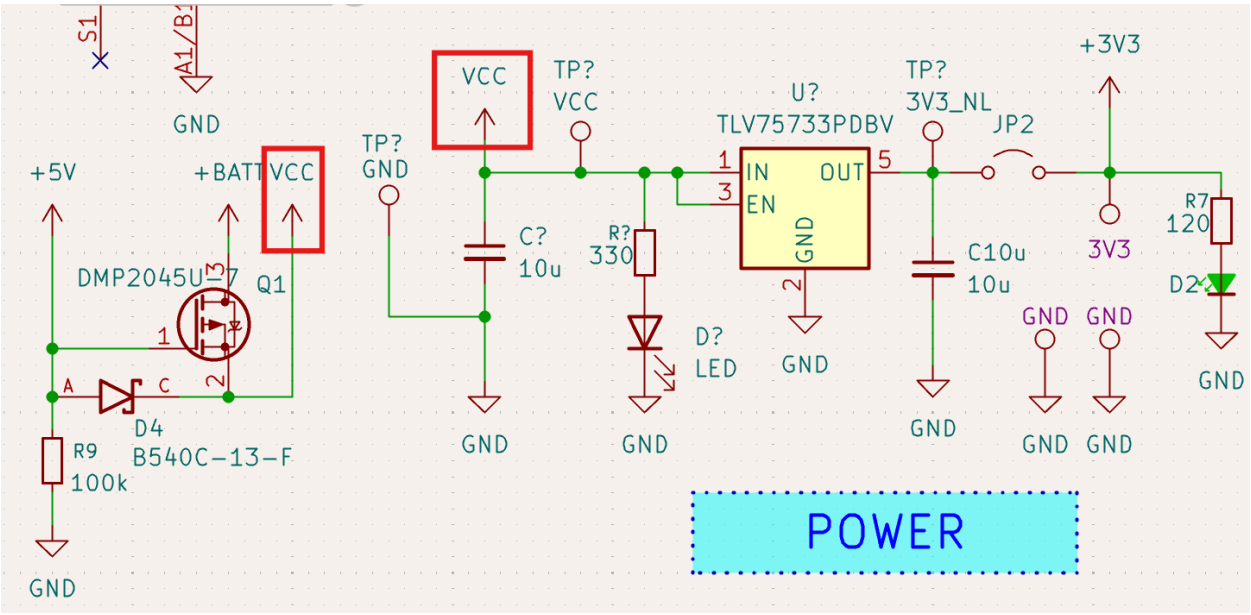


Figure 3: Example of power symbols. The VCC power symbols are highlighted here in red. They are connected to each other, and to the rest of the circuit, even though there is no wire running between them.

Below, Table 1 summarizes the reference designators (ref des), library locations, values, and footprints (physical realizations) of all of the component footprints you will need to add and update in your KiCad schematic file. Note that there are several additional components you will need to add to your circuit that were not found in DA1. These additional components include a status indicator LED and current limiting resistor as well as devices called test points. A test point is a designated spot on a PCB that is meant for diagnostic purposes. It provides a location where a probe can be connected to measure a voltage or current. Test points can be as simple as an exposed pad of copper or they can be physical parts that are soldered to a PCB. We will use small metal loops as our test points. You will solder these next week as part of DA3.

Table 1: Parameters of components to place

Component	Description	Library	Value/Part number	Footprint
Capacitor	Regulator input capacitor	Capacitor_SMD	10 µF	C_0805_2012Metric
Capacitor	Regulator output capacitor	Capacitor_SMD	10 µF	C_0805_2012Metric
Resistor	LED current limiting resistor	Resistor_SMD	300 Ω	R_0805_2012Metric
LED	Power input indicator LED	LED_SMD	n/a	LED_1206_3216Metric
TLV75733PDBV	Voltage regulator	Regulator_Linear -or- Your personal library	TLV75733PDBV	SOT-23-5
Test point	Test point, Vcc	Makerstock	S2751-46R	S2751-46R
Test point	Test point, 3V3_NL	Makerstock	S2751-46R	S2751-46R
Test point	Test point, GND	Makerstock	S2751-46R	S2751-46R

It is important to point out that you will need access to the Makerstock library of footprints. The Makerstock library has footprints for the majority of the parts that are stocked in the Maker-E database of components. See instructions above in the **More Resources that may be Helpful** section that will instruct you on how to access the Makerstock libraries.

In order to change an existing KiCad component footprint you will need to right click on the component and choose Properties (or hit the E key). Figure 4 shows where on the component interface to change the footprint.

When you have updated and placed all of the components from Table 1, double check your schematic against the example given in Figure 5. Do you have all of the same components? Are they wired correctly?

Now it is time to move on to laying out your PCB. Save your work.

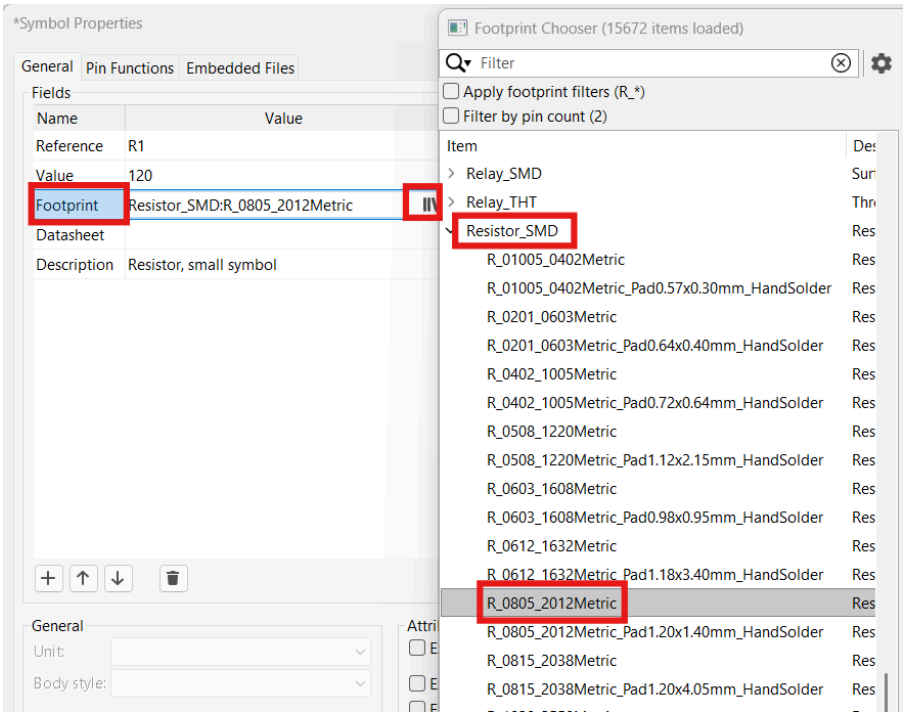


Figure 4: example of selecting a footprint for component R1

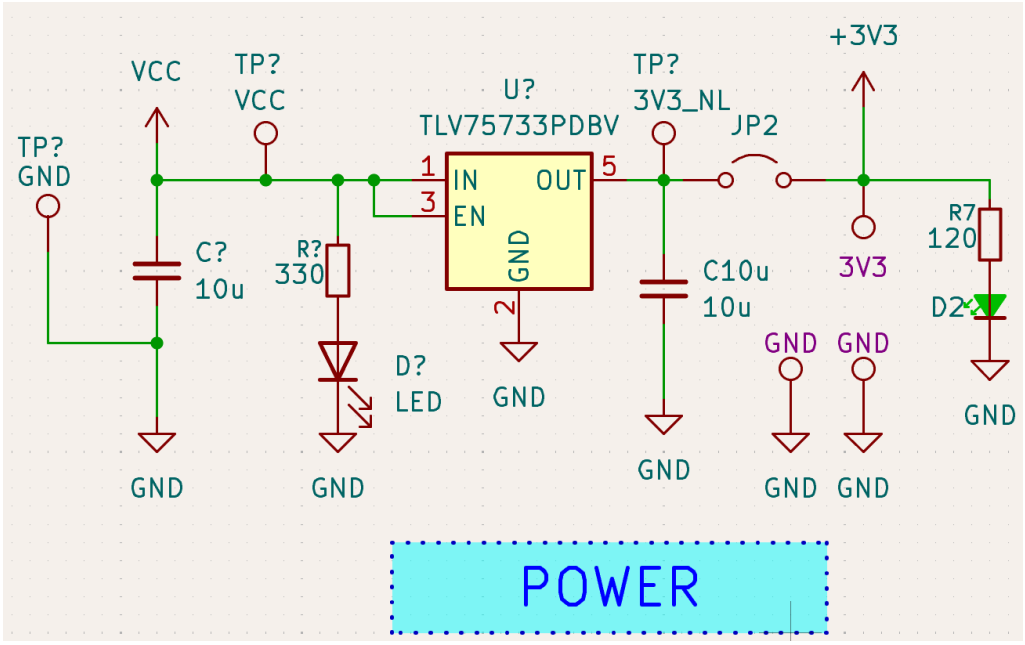


Figure 5: complete TLV757 LDO regulator circuit

PCB Editor

Now launch the KiCad PCB Editor and open the KiCad PCB file provided on Moodle. At first glance this may be overwhelming. It is a busy diagram with a lot going on. Recall that a PCB is composed of several layers. Each layer has its own color in KiCad. Here is a breakdown of the colors/layers:

- Grey: the perimeter rectangle is the outline of your board with a small rectangular cutout in the top
- Light blue: the criss-crossing lines are the “ratsnest.” These are virtual reminders only. They are a way of indicating to you that connections need to be made. It is up to you to run traces (wires) and make those connections. Once a connection is made, the ratsnest disappears.
- Red - top layer copper (wires and pads)
- Blue - bottom layer copper (wires and pads)
- Gold - copper on both top and bottom (typically surrounding through holes and vias)
- Light yellow - top layer silkscreen (human readable labels)
- Salmon - bottom layer silkscreen (human readable labels)
- Purple - indicates a locked item. Several items are locked since their position is crucial to board operation and they should not be moved.

Sometimes the default view can be overwhelming. In order to make this process easier you can selectively choose to hide certain layers to make the board appear less busy and cluttered. In the **Appearance** box at the top right, choose to hide these layers by clicking on the eye icon.

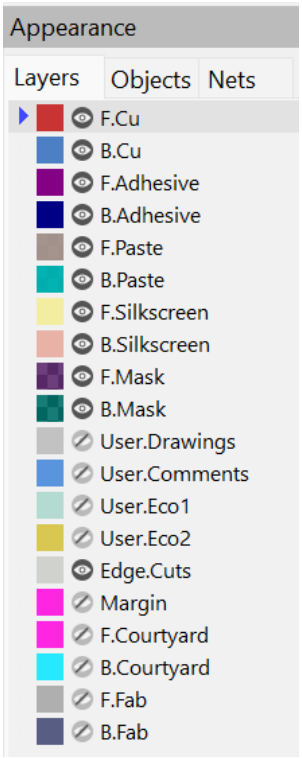


Figure 6: Hide these layers to clean up the view of the PCB. Clicking on the eye icon will place a slash across it and temporarily hide the layer from view.

Your task is now to take the information from the schematic, the components and how they are connected, and send that information into the PCB Editor. The board outline and most components are already in place in the PCB file. It is now up to you to move the new components for the voltage regulator module into place and connect them appropriately. The

connections you need to make will be shown via the light blue ratsnest lines. There are an infinite number of ways to place components and route traces to make these connections.

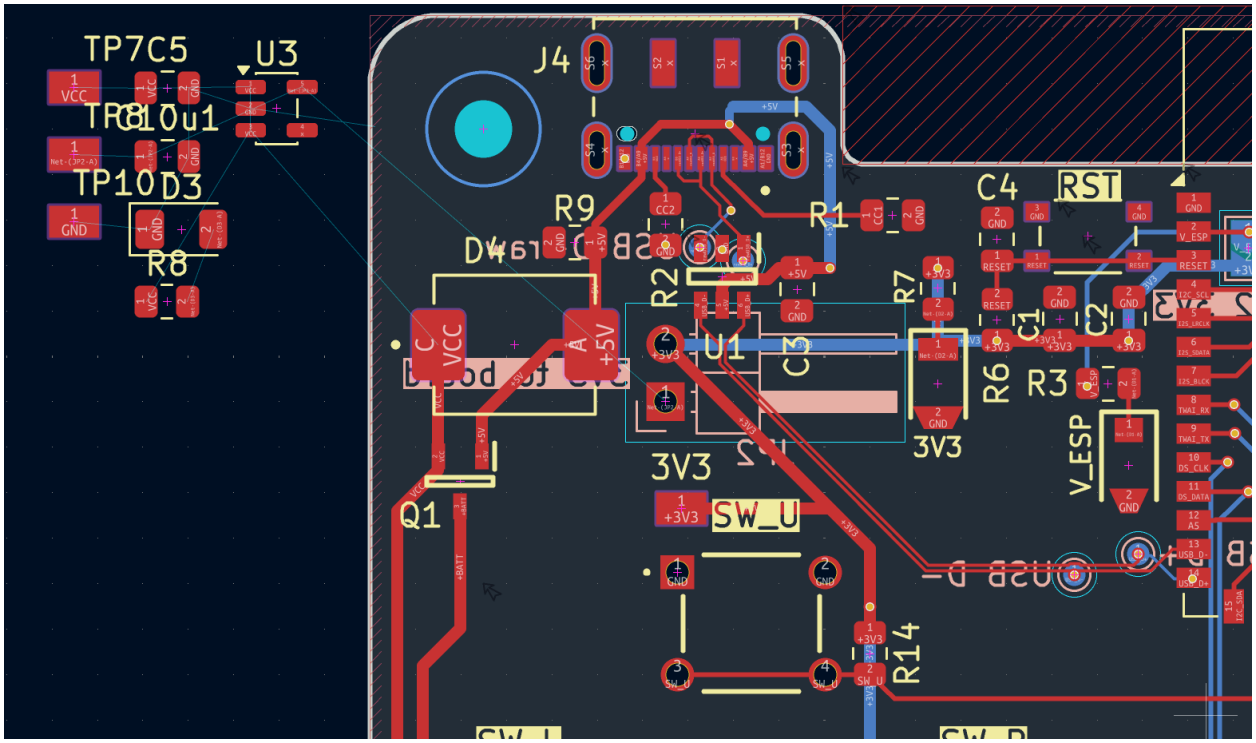


Figure 7: PCB with most components in place and new components at the top left awaiting placement.

You should read the KiCad Getting Started manual, beginning with the section, “Importing Changes From Schematic”. That will show you how to update your PCB with changes from the associated schematic. Once you have brought over all of your components into the PCB Editor (Figure 7) then you can begin rotating components and running traces. Pay attention to the “Placing Footprints” and “Routing Tracks” sections of the manual.

Most components have been placed for you already like the ESP32 module, 20 LEDs, and various headers. **Do not move these components**. Instead, move all of the components that appear outside of the large grey PCB outline to the inside of the large outline.

The best area for placement has been highlighted for you in green as shown below in Figure 8. This area is relatively free of existing components and traces and appears as empty black space. It may look like a small area but recall that these components are also very small and a good PCB design will tightly cluster associated components from a single block or module, with very short traces linking the components together.

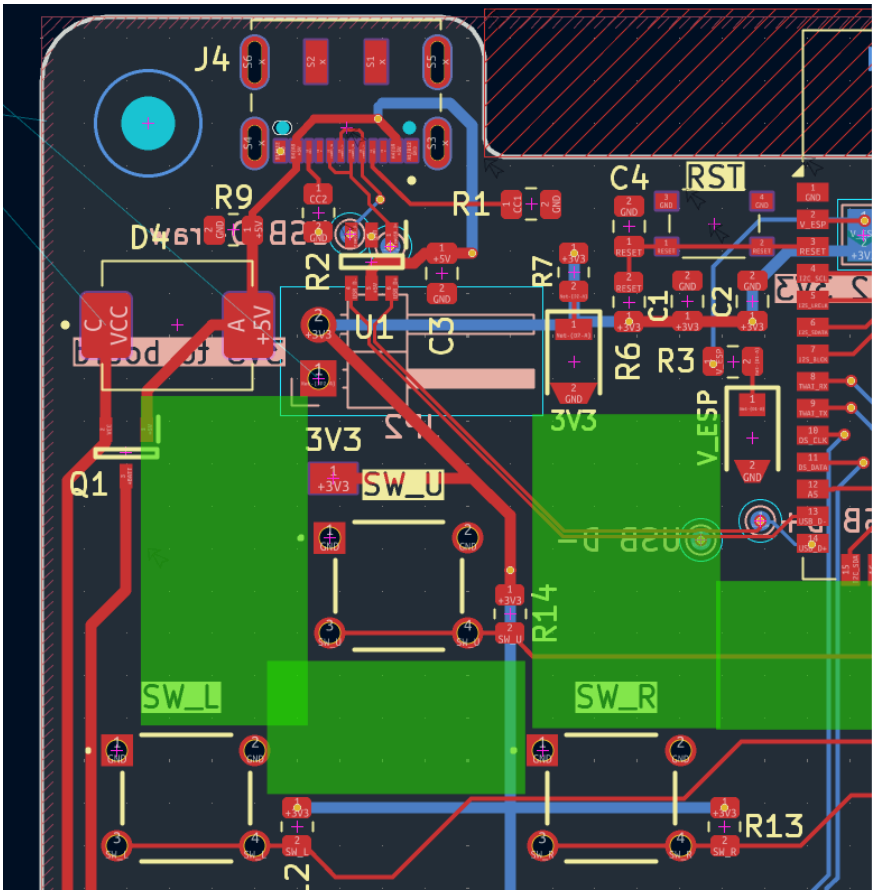


Figure 8: place your eight components for the voltage regulator circuit in the areas highlighted here in green

The goal is to minimize the amount of criss-crossing done by the light blue ratsnest lines before ever running any traces. In KiCad, the **R** key will rotate a component 90 degrees which is very helpful in minimizing ratsnest line crossing. This circuit is complex enough that it may not be possible to completely eliminate the crossing of ratsnest lines. But try to keep the ratsnest crossing to a minimum. When ratsnest lines cross, and you want to lay traces which cannot cross, you will have to place a via and move one trace to a different copper layer of the board - this is explained in the tutorial.

Once you are satisfied with your component placement then it is time to begin laying traces. You should use the **Route -> Route Single Track** command (**X** key) to place traces so you can avoid making incorrect connections. Remember that you can run traces on the top and bottom of your PCB. You can switch between layers (**V** key) by connecting traces together with vias (CTRL + shift + **X**). Just keep in mind that surface mount (SMD) components **must** be connected to traces that are on the same side as the component itself is mounted. In our case all of the SMD components are on the top layer, so all SMD traces should begin on the top layer in order to connect to the SMD component pads. Once a trace is connected to a top layer pad you may then switch to the bottom layer using a via.

This part may take you a while. You may get halfway done running traces and hit a wall where you realize you cannot go any farther. In this case you may need to delete a few traces and move a few components before you can eliminate the problem and continue working. Don't lose hope! Talk to your classmates and any upper class folks you know who have created PCBs already. Third year students who took this class last year are an excellent resource! Feel free to share ideas on how best to place and orient components. While everyone will be designing the same board, with the same components, no two PCBs should look the same.

When you are done placing components and routing traces, run the Design Rule Checker (DRC) from the Inspect menu. You may get lots of warnings and errors. You can uncheck the box for warnings. You can also ignore the errors. What we want to focus on is the **Unconnected Items** tab. This is the area where KiCad will inform you if you forgot to connect any components and there are remaining ratsnest connections which you need to resolve. Double click on any violations in the Unconnected Items tab and the PCB Editor will zoom in to show you where the missing connection lies. For the purpose of this assignment I am not concerned with board errors. However, you **do** need to make sure that you make all connections. This means that there should be no items remaining in the Unconnected Items tab.

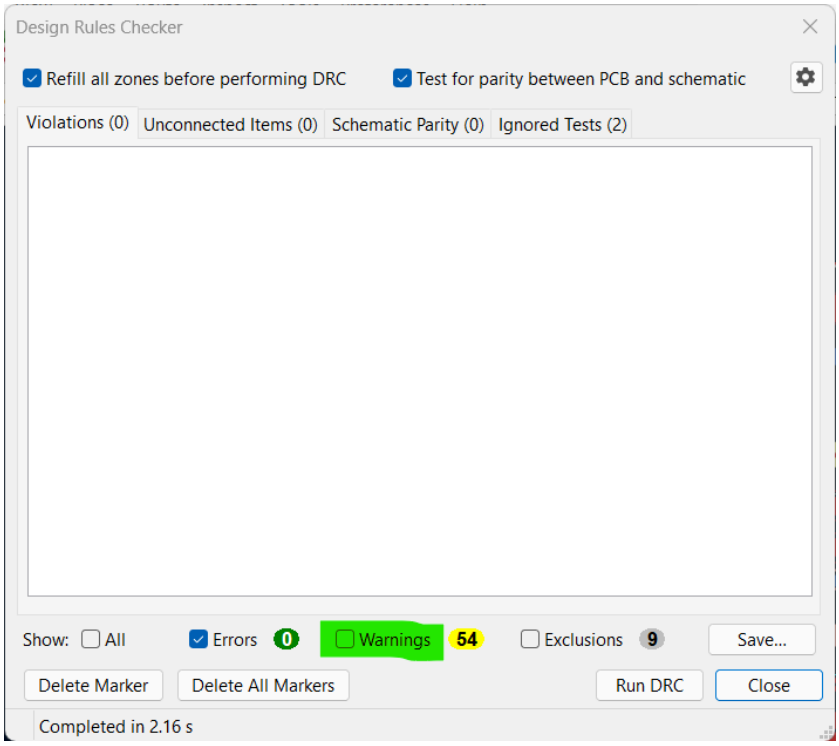


Figure 9: example result of DRC with Warnings box unchecked and no remaining unconnected items

When designing your own boards you will want to carefully review all DRC warnings and errors as these indicate that design rules used for fabrication have been violated which means your

board will not be manufactured properly. However, for this design assignment we can ignore all errors and warnings. You are only responsible for any items listed in the Unconnected Items tab. When you are done:

- All of your components should be within the bounds of the PCB (grey rectangle).
- There should be no ratsnest connections (straight light blue lines connecting components) remaining; in other words you should have run traces to make all of the required component connections between pins. You can check this using the Design Rule Checker as outlined above.
- No component should overlap with another component. This only applies to the silkscreen layer and component pads. Traces can overlap, as long as they are on separate layers.

A word of warning: as you lay traces in this design there is the possibility that **new** ratsnest connections may appear that previously were not present. This is because there are two hidden copper planes in use on this PCB. These large swaths of copper, or power planes, are used for ground in our circuit. [Such ground planes have other advantages in circuit design. They help shield the circuit from picking up electrical noise from the environment and can help high frequency signals propagate between points on the printed circuit board. You'll learn about high frequency signal propagation in ECEG 390 and noise in ECEG 370]. Because so many devices need connections to power and ground it is common practice to create entire copper layers dedicated to these nets. As you lay traces on a layer, and that hidden copper is divided by your traces, you may create copper "islands". If you are interested in learning more about this you can check out this informative video about copper islands. You can also temporarily enable the viewing of these copper areas by clicking the **Show filled areas of zones** button. After moving or rotating components, be sure to hit the **B** key on your keyboard to refill the zones and update these copper areas. The take home is this - always double check your PCB when you are done and confirm you have no ratsnest connections remaining. Check out the Placing Copper Zones section of the Getting Started guide for more information.

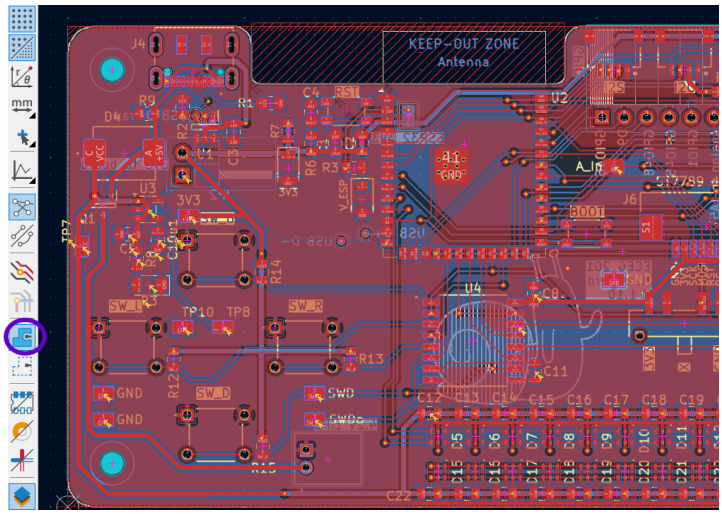


Figure 10: filled copper areas shown. View enabled by clicking the button circled here in purple.

Be prepared to spend some time on this design. It is very likely that you will need to remove components and traces more than once in order to squeeze everything in and wire it correctly. Once you are satisfied with your PCB design be sure to save your work.

Once you've got your design finalized you'll need to add dimensions. Adding dimensions is not necessary for having your board manufactured. However, if you need to share your board design with another engineer who is designing an enclosure or any other devices that will interface with your PCB, then creating a technical drawing with dimensions is an important way to share information. You need to add three dimensions:

- Outer PCB dimensions (120 mm x 55 mm)
- Center to center distance between mounting holes (110 mm and 45 mm)
- Diameter of the four mounting holes (2.5 mm)

Begin by activating the layer upon which you wish to place the dimensions. For our purposes we will place dimensions on the User.1 layer. This layer is used purely for informational purposes and will have no effect on board production. Be sure to enable viewing of this layer if you chose to hide it previously. Click on this layer in the right **Appearance** window to activate it. You should also make your grid relatively large so the cursor will only snap to reasonable points on the board. Choose **1.00 mm** from the dropdown menu near the top center of the screen.

Now go to **Place -> Draw Dimensions -> Draw Orthogonal Dimension** from the top menu bar. Zoom in on the bottom left edge of the board where switches SW_L and SW_D are located. Be sure to click on the bottom left edge of the board (where the origin crosshairs indicator is).

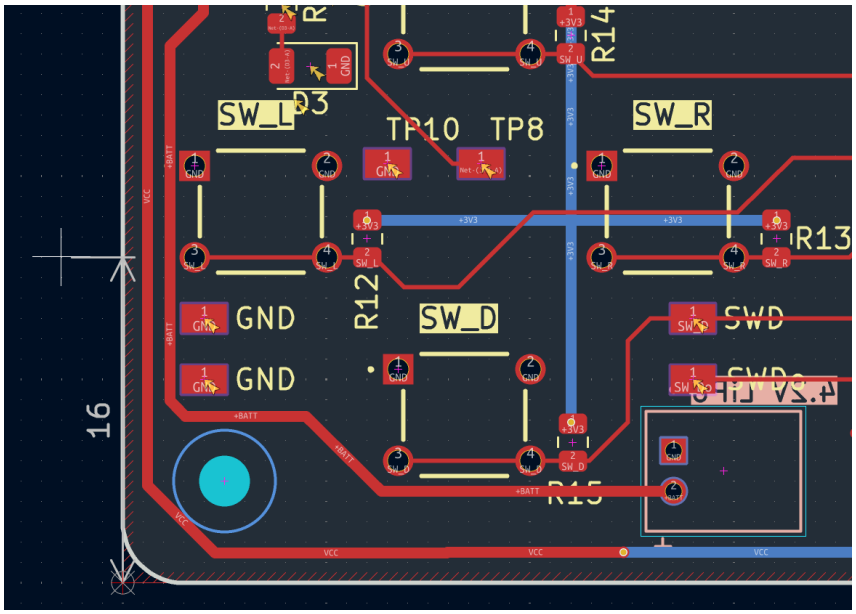


Figure 11: Beginning to place the vertical board dimension by starting at the bottom left corner of the board (board origin crosshairs)

Zoom out and click a second time on the top left corner of the board. You should then be presented with a set of arrows with a measurement between them. Drag this measurement left, beyond the board and click one more time to anchor it. The value between the arrows should read 55.

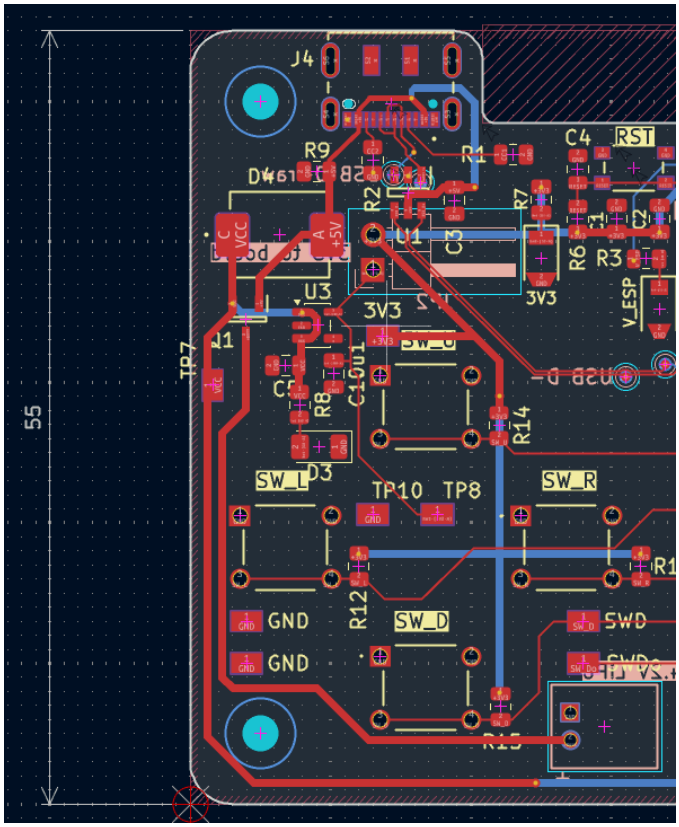


Figure 12: correct placement and measurement of PCB vertical dimension

Now repeat this procedure for the bottom of the board and place a dimension for the horizontal dimension (120 mm).

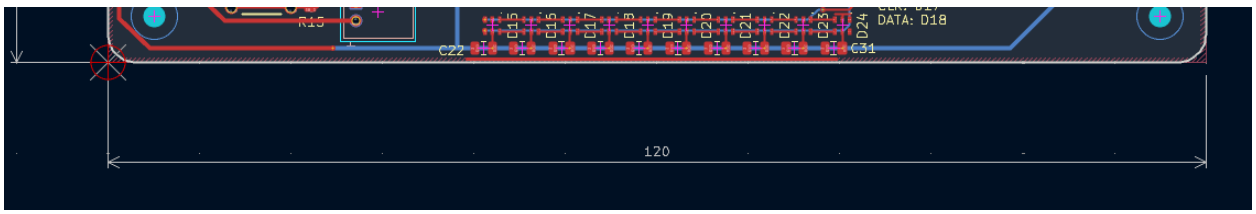


Figure 13: correct placement and measurement of PCB horizontal dimension

Again repeat this process but this time choose **Place -> Draw Dimensions -> Draw Aligned Dimension** from the top menu bar. Click on the center of the mounting hole at the bottom left of the PCB. Then click on the center of the bottom right mounting hole. This provides the center-to-center distance for these holes which is useful information when designing the

enclosure for this board. Repeat this process for vertical mounting holes on the left side of the board. When you're done placing dimensions your screen should look like this.

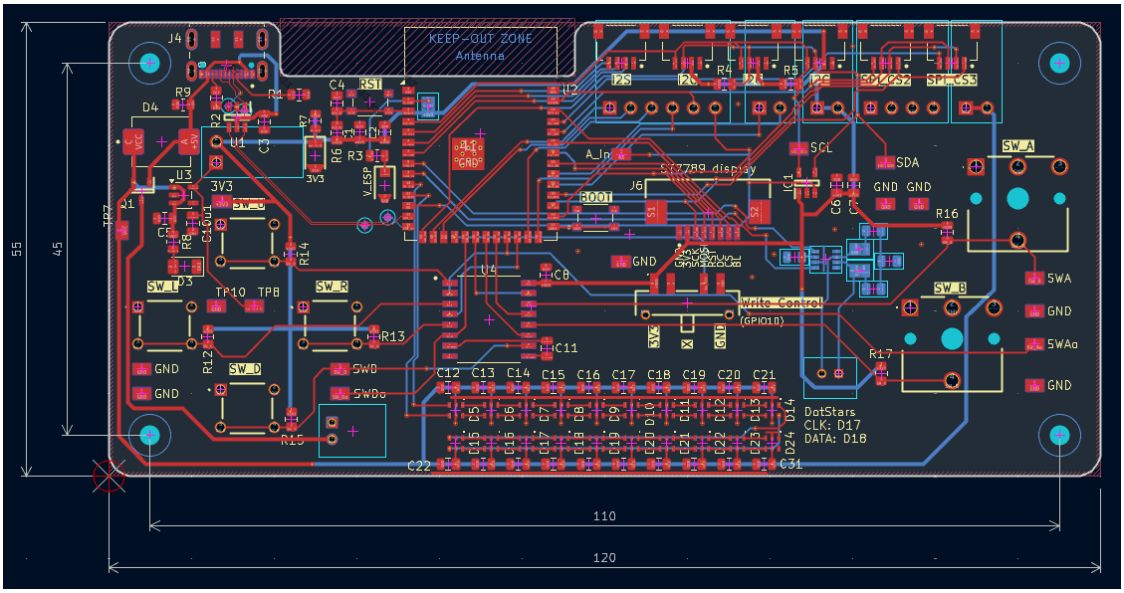


Figure 14: four correct dimensions placed

Finally, we need to add a dimension to indicate the size of the mounting holes. Choose **Place -> Draw Dimensions -> Draw Leaders** from the top menu bar. Click on the center of the top left mounting hole and then move the resultant measurement out away from the board to an easy to view position. When prompted to enter text you should enter the standard CAD notation for hole diameters: Ø 2.5 mm. Once you have all of your dimensions your board should look like the figure below.

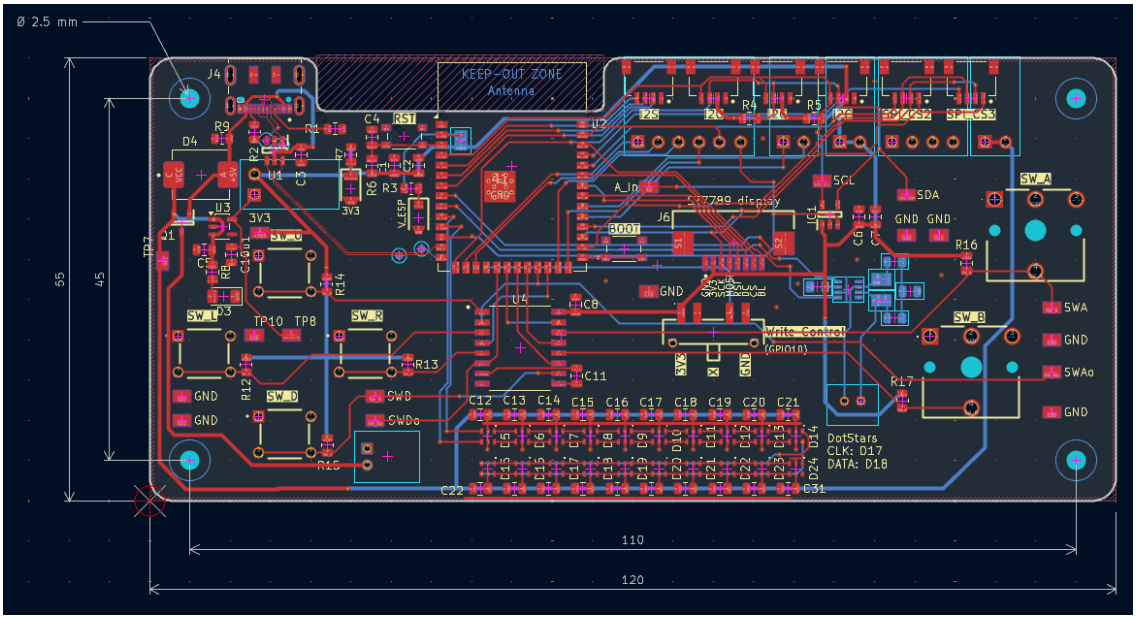


Figure 15: complete set of dimensions added to PCB

Once you are satisfied with your dimensions you should print a copy of the board as a PDF file to be used as a dimensional drawing. Choose **File -> Print** and then use the following settings in the print dialog window:

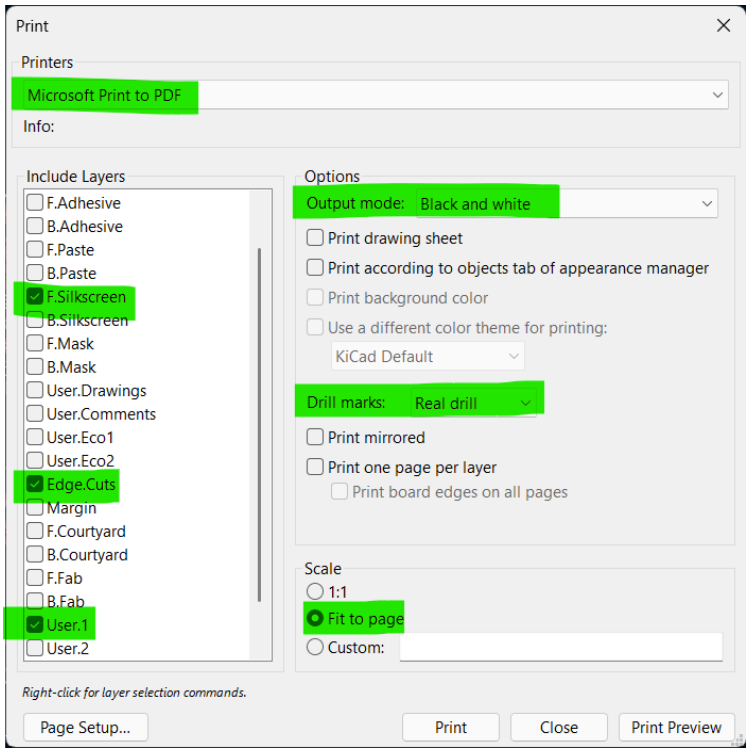


Figure 16: Print settings to be used for PDF export

Click on **Print** and choose to print the board as a PDF. Save the PDF somewhere you will be able to access it later (like in your Google Drive folder for DA2). If you do not have the option to save as a PDF, talk to Matt for an alternative.

What to Submit

You should submit required documents in your course Google Drive folder. Be sure to create a separate DA2 folder for this assignment. The link to your Google Drive folder can be found at the top of the course Moodle page. For this assignment you should submit your three KiCad files: project, schematic, PCB. You should also update Section 9.1 “Printed Circuit Board” of your DAMNED Datasheet template.

First, make a copy of your DAMNED datasheet template that you submitted for DA1. Be sure to remove highlighting from DA1, but do not remove the information you added for DA1. The idea here is that we will build up the datasheet template, one section at a time, as you complete each design assignment. So each week you will copy the previous week’s template, update the document name to reflect the new DA number, and then add the requested information from the current DA. Make sure you place a completed PDF copy of the datasheet template in your

Google Drive folder.

The only thing you need to add to Section 9.1 for DA2 is a copy of the dimensional drawing you created in the PDF above.